Guided Tutorial on Logic

This guided tutorial is designed to help you understand how logic works in Artificial Intelligence, particularly propositional logic, inference techniques, and predicate logic. We will explore these concepts using clear examples, truth tables, and reasoning steps.

1. Key Definitions (In Simple Terms)

Term	Meaning
Propositional Logic	A formal system where statements (propositions) are either true or false.
Predicate Logic	An extension of propositional logic that includes variables and quantifiers.
Entailment	If a knowledge base (KB) implies a statement logically; i.e., KB $\models \alpha$.
Model	An interpretation that defines the truth of each sentence.
Inference	The process of deriving new facts from known ones.
Validity	A sentence true in all models.
Satisfiability	A sentence that is true in at least one model.

2. Propositional Logic

Propositional logic represents facts as symbols such as P, Q, and R. These can be combined with logical connectives to express complex sentences.

Common Connectives: AND ($\wedge$), OR ($\vee$), NOT ($\neg$), IMPLIES ($\Rightarrow$), IFF ($\Leftrightarrow$).

Example:
"If it rains, the grass is wet" $\rightarrow R \Rightarrow W$
"It rains" $\rightarrow R$
Therefore, "The grass is wet" $\rightarrow W$

Example:

Truth Table for $(P \vee Q) \Rightarrow R$

P	Q	R	P ∨Q	(P ∨Q) ⇒R
F	F	F	F	T
F	F	T	F	T
F	T	F	T	F
F	T	T	T	T
T	F	F	T	F
T	F	T	T	T
T	T	F	T	F
T	T	T	T	T

Interpretation:

- The only time (P∨Q)⇒R is false is when P∨Q is true but R is false.
- This table clearly shows the dependency of implication on the truth of the antecedent (P ∨ Q) and the consequent (R).

3. Inference and Proof Methods

Inference allows us to draw conclusions from existing facts using sound rules of logic. These methods form the foundation for reasoning systems and automated theorem proving.

Key Inference Rules and Examples:

Modus Ponens: If P ⇒ Q and P are true, then Q must be true.
Example: If it is raining, the ground is wet. It is raining ⟹ The ground is wet.

Modus Tollens: If P ⇒ Q and ¬Q are true, then ¬P is true.
Example: If it rains, the ground is wet. The ground is not wet ⟹ It is not raining.

Disjunctive Syllogism: If P ∨ Q and ¬P are true, then Q is true.
Example: It is raining or snowing. It is not raining ⟹ It is snowing.

Hypothetical Syllogism: If P ⇒ Q and Q ⇒ R, then P ⇒ R.
Example: If I study, I pass. If I pass, I graduate ⟹ If I study, I graduate.

Conjunction: If P and Q are both true, then P ∧ Q is true.
Example: It is sunny and warm ⟹ True.

Simplification: If P ∧ Q is true, then P is true.
Example: It is sunny and warm ⟹ It is sunny.

Addition: If P is true, then P ∨ Q is true.
Example: It is sunny ⟹ It is sunny or cloudy.

4. Forward and Backward Chaining

Forward chaining is data-driven, starting from facts and applying rules to infer new facts until the goal is reached. Backward chaining is goal-driven, starting from the goal and working backward to find the supporting facts.

Forward Chaining: https://www.youtube.com/watch?v=Dt2pNoTIcMc

Backward Chaining: https://www.youtube.com/watch?v=OYqz-_YbWbU

Example of symbolic forward chaining:

1. $A \land B \Rightarrow C$
2. $C \land D \Rightarrow E$
3. $E \lor F \Rightarrow G$
4. $G \Rightarrow H$

Facts: A, B, D are true.
$\Rightarrow C \Rightarrow E \Rightarrow G \Rightarrow H$

Example of backward chaining:

Goal: Prove H.
Rules:
1. $G \Rightarrow H$
2. $E \lor F \Rightarrow G$
3. $C \land D \Rightarrow E$
4. $A \land B \Rightarrow C$

Facts: A, B, D are true.
To prove H $\Rightarrow$ need G $\Rightarrow$ need $E \lor F \Rightarrow$ need $C \land D \Rightarrow$ need $A \land B \Rightarrow$ all true $\Rightarrow$ H proven.

5. Resolution

Resolution is a rule of inference used for automated theorem proving. It shows inconsistency between a KB and the negation of a query (proof by contradiction).

Proof by Resolution: https://www.youtube.com/watch?v=nEEyPdYxBFY

Example:
1. $A \vee B$
2. $\neg A \vee C$
3. $\neg B \vee C$

Goal: Prove C.
Resolving (1) and (2): $B \vee C$
Resolving with (3): $C \Longrightarrow$ Proven.

6. First-Order Logic (Predicate Logic)

First-order logic extends propositional logic by using **quantifiers**, **variables**, and **predicates** to express relations among objects.

Basic Components:

- **Constants:** KingJohn, Socrates, Paris
- **Predicates:** Human(x), Loves(x, y), GreaterThan(a, b)
- **Quantifiers:** ∀ (for all), ∃ (there exists)

Example: Universal Quantifier

$\forall x$ (Human(x) $\Rightarrow$ Mortal(x))
"All humans are mortal."

Example: Existential Quantifier

$\exists x$ (Student(x) $\wedge$ StudiesAI(x))
"There exists a student who studies AI."

Example:

$\forall x \forall y$ (Parent(x, y) $\Rightarrow$ Ancestor(x, y))
$\forall x \forall y \forall z$ ((Parent(x, y) $\wedge$ Ancestor(y, z)) $\Rightarrow$ Ancestor(x, z))
"If x is a parent of y and y is an ancestor of z, then x is also an ancestor of z."

Example:

1. $\forall x$ (Employee(x) $\wedge$ WorksFor(x, CompanyA) $\Rightarrow$ HasID(x))
2. $\forall x$ (Employee(x) $\wedge$ Manager(x) $\Rightarrow$ AccessLevelHigh(x))
3. Employee(John), WorksFor(John, CompanyA), Manager(John)

$\Longrightarrow$ HasID(John) $\wedge$ AccessLevelHigh(John)

7. Comparison of Logics

Feature	Propositional Logic	Predicate Logic
Expressiveness	Low	High
Uses Quantifiers	No	Yes
Inference Complexity	Low	High
Use Case	Simple rule-based reasoning	Knowledge representation

8. Summary

Logic provides the foundation for reasoning in AI. Propositional logic handles simple facts, while predicate logic allows more complex relationships. Inference mechanisms like chaining and resolution automate reasoning.

9. Try It Yourself (Exercises)

1. Construct a truth table for $(P \Rightarrow Q) \land (Q \Rightarrow R)$.
2. Use resolution to prove $(P \Rightarrow Q), (Q \Rightarrow R), P \vDash R$.
3. Express in FOL: 'Every student attends at least one lecture.'
4. Using forward chaining, infer 'It is snowing' given: $C \Rightarrow S$, $K \Rightarrow C$, K.
5. Translate 'No cat is a dog' into predicate logic.